

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

In [2]:

```
dataset=pd.read_csv('Mall_Customers.csv')
```

In [3]:

```
dataset
```

Out[3]:

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [4]:

```
x=dataset.iloc[:,3:5].values
```

In [5]:

```
x
```

Out[5]:

```
array([[ 15,  39],
       [ 15,  81],
       [ 16,   6],
       [ 16,  77],
       [ 17,  40],
       [ 17,  76],
       [ 18,   6],
       [ 18,  94],
       [ 19,   3],
       [ 19,  72],
       [ 19,  14],
       [ 19,  99],
       [ 20,  15],
       [ 20,  77],
       [ 20,  13],
```

[20, 79],
[21, 35],
[21, 66],
[23, 29],
[23, 98],
[24, 35],
[24, 73],
[25, 5],
[25, 73],
[28, 14],
[28, 82],
[28, 32],
[28, 61],
[29, 31],
[29, 87],
[30, 4],
[30, 73],
[33, 4],
[33, 92],
[33, 14],
[33, 81],
[34, 17],
[34, 73],
[37, 26],
[37, 75],
[38, 35],
[38, 92],
[39, 36],
[39, 61],
[39, 28],
[39, 65],
[40, 55],
[40, 47],
[40, 42],
[40, 42],
[42, 52],
[42, 60],
[43, 54],
[43, 60],
[43, 45],
[43, 41],
[44, 50],
[44, 46],
[46, 51],
[46, 46],
[46, 56],
[46, 55],
[47, 52],
[47, 59],
[48, 51],
[48, 59],
[48, 50],
[48, 48],
[48, 59],
[48, 47],
[49, 55],
[49, 42],
[50, 49],
[50, 56],

[54, 47],
[54, 54],
[54, 53],
[54, 48],
[54, 52],
[54, 42],
[54, 51],
[54, 55],
[54, 41],
[54, 44],
[54, 57],
[54, 46],
[57, 58],
[57, 55],
[58, 60],
[58, 46],
[59, 55],
[59, 41],
[60, 49],
[60, 40],
[60, 42],
[60, 52],
[60, 47],
[60, 50],
[61, 42],
[61, 49],
[62, 41],
[62, 48],
[62, 59],
[62, 55],
[62, 56],
[62, 42],
[63, 50],
[63, 46],
[63, 43],
[63, 48],
[63, 52],
[63, 54],
[64, 42],
[64, 46],
[65, 48],
[65, 50],
[65, 43],
[65, 59],
[67, 43],
[67, 57],
[67, 56],
[67, 40],
[69, 58],
[69, 91],
[70, 29],
[70, 77],
[71, 35],
[71, 95],
[71, 11],
[71, 75],
[71, 9],
[71, 75],
[72, 34],

[72, 71],
[73, 5],
[73, 88],
[73, 7],
[73, 73],
[74, 10],
[74, 72],
[75, 5],
[75, 93],
[76, 40],
[76, 87],
[77, 12],
[77, 97],
[77, 36],
[77, 74],
[78, 22],
[78, 90],
[78, 17],
[78, 88],
[78, 20],
[78, 76],
[78, 16],
[78, 89],
[78, 1],
[78, 78],
[78, 1],
[78, 73],
[79, 35],
[79, 83],
[81, 5],
[81, 93],
[85, 26],
[85, 75],
[86, 20],
[86, 95],
[87, 27],
[87, 63],
[87, 13],
[87, 75],
[87, 10],
[87, 92],
[88, 13],
[88, 86],
[88, 15],
[88, 69],
[93, 14],
[93, 90],
[97, 32],
[97, 86],
[98, 15],
[98, 88],
[99, 39],
[99, 97],
[101, 24],
[101, 68],
[103, 17],
[103, 85],
[103, 23],
[103, 69],

```
[113, 8],  
[113, 91],  
[120, 16],  
[120, 79],  
[126, 28],  
[126, 74],  
[137, 18],  
[137, 83]])
```

In [9]:

```
from sklearn.cluster import KMeans  
list1 = []  
for i in range(1, 11):  
    kmeans = KMeans(n_clusters = i, init = 'k-means++', random_state = 42)  
    kmeans.fit(x)  
    list1.append(kmeans.inertia_)  
plt.plot(range(1, 11), list1)  
plt.title('The Elbow Method')  
plt.xlabel('Number of clusters')  
plt.ylabel('WCSS')  
plt.show()
```

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

warnings.warn(

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

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C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

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C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

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C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

warnings.warn(

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

warnings.warn(

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

warnings.warn(

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

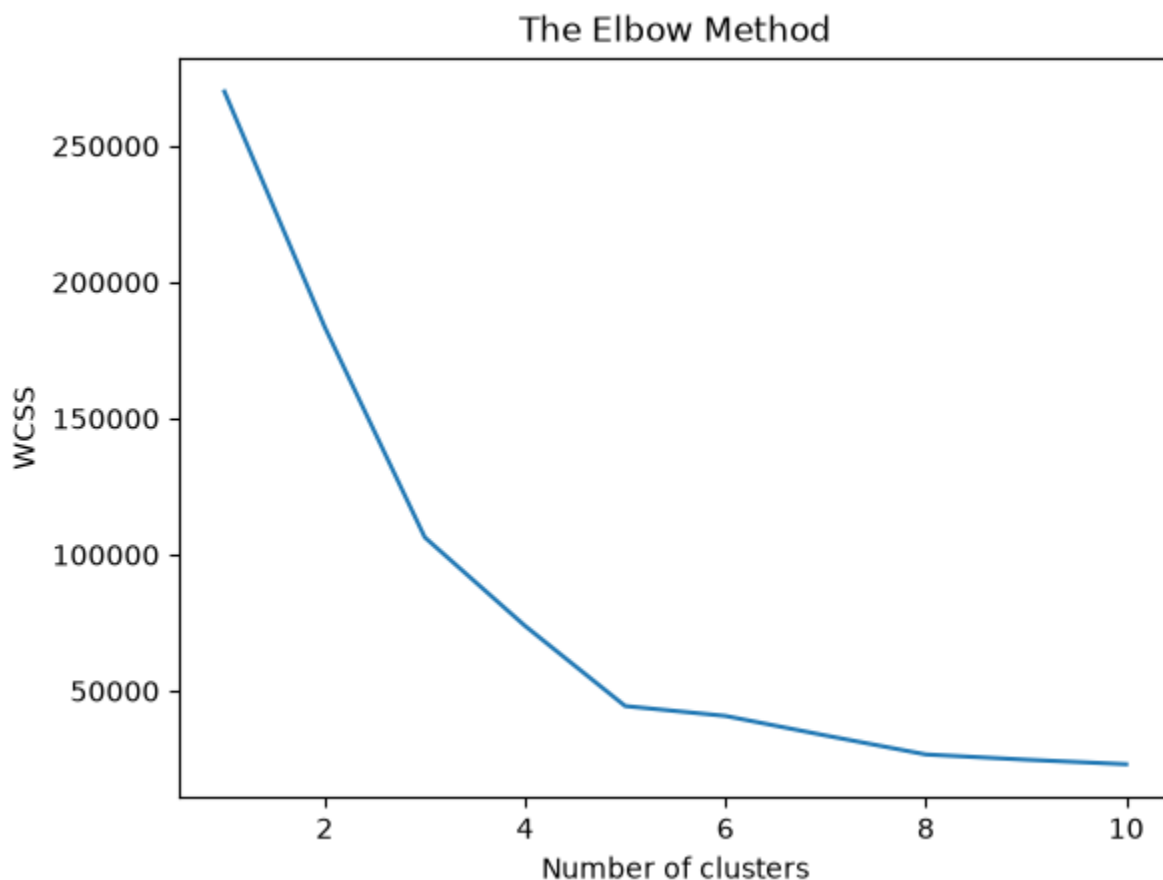
an available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

```
warnings.warn(  
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserWarning:  
KMeans is known to have a memory leak on Windows with MKL, when there are less chunks th  
an available threads. You can avoid it by setting the environment variable OMP_NUM_THREA  
DS=1.
```

```
warnings.warn(  
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1428: UserWarning:  
KMeans is known to have a memory leak on Windows with MKL, when there are less chunks th  
an available threads. You can avoid it by setting the environment variable OMP_NUM_THREA  
DS=1.
```

```
warnings.warn(  

```



In [10]:

```
list1
```

Out[10]:

```
[269981.28,  
183653.32894736846,  
106348.3730621112,  
73880.64496247195,  
44448.45544793371,  
40825.16946386947,  
33642.57922077922,  
26686.83778518779,  
24766.47160979344,  
23103.122085983912]
```

In [11]:

```
from sklearn.cluster import KMeans  
kmeans = KMeans(n_clusters = 5, init = 'k-means++', random_state = 42)
```

```
y_kmeans = kmeans.fit_predict(x)
```

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1428: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

```
warnings.warn(
```

In [12]:

```
y_kmeans
```

Out[12]:

```
array([4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2,
        4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2,
        4, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 0, 1, 3, 1, 3, 1, 3, 1, 3, 1,
        3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1,
        3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1,
        3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1,
        3, 1], dtype=int32)
```

In [13]:

```
supervised=dataset
```

In [14]:

```
supervised
```

Out[14]:

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [15]:

```
supervised['Cluster_group']=y_kmeans
```

In [16]:

```
supervised
```

Out[16]:

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	Cluster_group
0	1	Male	19	15	39	4
1	2	Male	21	15	81	2
2	3	Female	20	16	6	4
3	4	Female	23	16	77	2
4	5	Female	31	17	40	4
...	...	...	...	...	...	...
195	196	Female	35	120	79	1
196	197	Female	45	126	28	3
197	198	Male	32	126	74	1
198	199	Male	32	137	18	3
199	200	Male	30	137	83	1

200 rows × 6 columns

In [17]:

```
supervised.to_csv("cluster.csv", index=False)
```

In [18]:

```
dir(kmeans)
```

Out[18]:

```
['__abstractmethods__',
 '__annotate_func__',
 '__class__',
 '__delattr__',
 '__dict__',
 '__dir__',
 '__doc__',
 '__eq__',
 '__firstlineno__',
 '__format__',
 '__ge__',
 '__getattr__',
 '__getstate__',
 '__gt__',
 '__hash__',
 '__init__',
 '__init_subclass__',
 '__le__',
 '__lt__',
 '__module__',
 '__ne__',
 '__new__',
 '__reduce__',
 '__reduce_ex__',
 '__repr__',
 '__setattr__',
 '__setstate__',
 '__sizeof__',
```

```
'__sklearn_clone__',
'__sklearn_tags__',
'__slots__',
'__static_attributes__',
'__str__',
'__subclasshook__',
'__weakref__',
'_abc_impl',
'_algorithm',
'_check_mkl_vcomp',
'_check_params_vs_input',
'_check_test_data',
'_doc_link_module',
'_doc_link_template',
'_doc_link_url_param_generator',
'_get_class_level_metadata_request_values',
'_get_doc_link',
'_get_fitted_attr_html',
'_get_metadata_request',
'_get_param_names',
'_get_params_html',
'_html_repr',
'_init_centroids',
'_n_features_out',
'_n_init',
'_n_threads',
'_parameter_constraints',
'_repr_html_',
'_repr_html_inner',
'_repr_mimebundle_',
'_sklearn_auto_wrap_output_keys',
'_tol',
'_transform',
'_validate_center_shape',
'_validate_params',
'_warn_mkl_vcomp',
'algorithm',
'cluster_centers_',
'copy_x',
'fit',
'fit_predict',
'fit_transform',
'get_feature_names_out',
'get_metadata_routing',
'get_params',
'inertia_',
'init',
'labels_',
'max_iter',
'n_clusters',
'n_features_in_',
'n_init',
'n_iter_',
'predict',
'random_state',
'score',
'set_fit_request',
'set_output',
'set_params',
```

```
'set_score_request',  
'tol',  
'transform',  
'verbose']
```

In [19]:

```
centroids=kmeans.cluster_centers_
```

In [20]:

```
centroids
```

Out[20]:

```
array([[55.2962963 , 49.51851852],  
       [86.53846154, 82.12820513],  
       [25.72727273, 79.36363636],  
       [88.2         , 17.11428571],  
       [26.30434783, 20.91304348]])
```

In [21]:

```
y_kmeans
```

Out[21]:

```
array([4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2,  
       4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 0,  
       4, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 3, 1, 0, 1, 3,  
       0, 1, 3, 1, 3, 1, 3, 1, 3, 1, 0, 1, 3, 1, 3, 1, 3, 1, 3, 1,  
       3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1,  
       3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1,  
       3, 1], dtype=int32)
```

In [22]:

```
supervised.columns[3]
```

Out[22]:

```
'Annual Income (k$)'
```

In [23]:

```
supervised.columns
```

Out[23]:

```
Index(['CustomerID', 'Genre', 'Age', 'Annual Income (k$)',  
       'Spending Score (1-100)', 'Cluster_group'],  
      dtype='str')
```

In [35]:

```
!pip install seaborn
```

```
Defaulting to user installation because normal site-packages is not writeable  
Requirement already satisfied: seaborn in C:\ProgramData\anaconda3\Lib\site-packages (0.  
13.2)  
Requirement already satisfied: numpy!=1.24.0,>=1.20 in C:\ProgramData\anaconda3\Lib\site-  
packages (from seaborn) (2.4.6)  
Requirement already satisfied: pandas>=1.2 in C:\ProgramData\anaconda3\Lib\site-packages  
(from seaborn) (3.0.3)  
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in C:\ProgramData\anaconda3\Lib\s  
ite-packages (from seaborn) (3.11.0)  
Requirement already satisfied: contourpy>=1.0.1 in C:\ProgramData\anaconda3\Lib\site-pac  
kages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
```

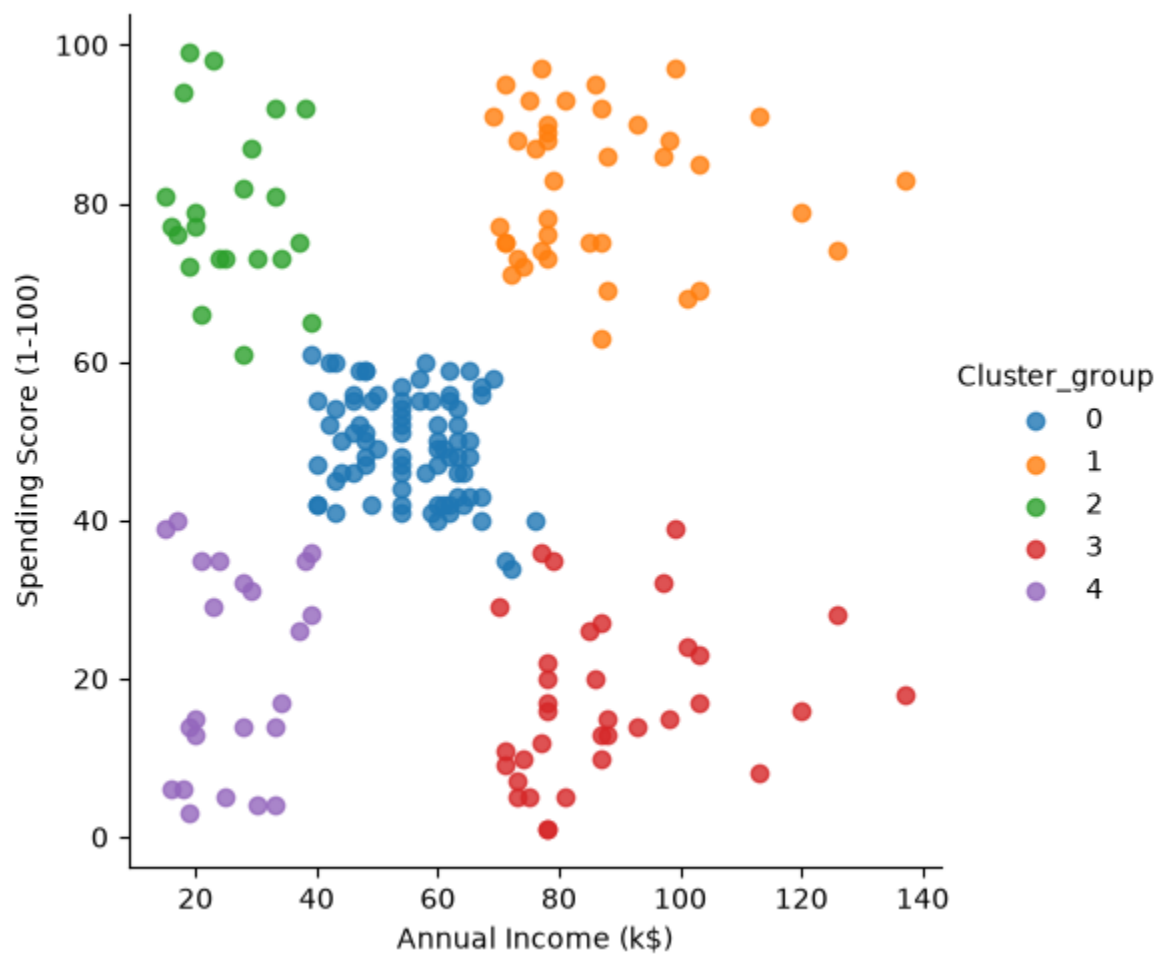
Requirement already satisfied: cycler>=0.10 in C:\ProgramData\anaconda3\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in C:\ProgramData\anaconda3\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\ProgramData\anaconda3\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)
Requirement already satisfied: packaging>=20.0 in C:\ProgramData\anaconda3\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.0)
Requirement already satisfied: pillow>=9 in C:\ProgramData\anaconda3\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\ProgramData\anaconda3\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in C:\ProgramData\anaconda3\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\ProgramData\anaconda3\Lib\site-packages (from pandas>=1.2->seaborn) (2026.2)
Requirement already satisfied: six>=1.5 in C:\ProgramData\anaconda3\Lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)

In [39]:

```
import seaborn as sns

facet = sns.lmplot(
    data=supervised,
    x=supervised.columns[3],
    y=supervised.columns[4], hue=supervised.columns[5],
    fit_reg=False,
    legend_out=True
)
```

C:\ProgramData\anaconda3\Lib\site-packages\seaborn\regression.py:598: UserWarning: legend_out is deprecated from the `lmplot` function signature. Please update your code to pass it using `facet\_kws`.
warnings.warn(msg, UserWarning)



In []: